

Tezpur University

Autumn End Semester Examination–2025

Course Code: CS513**Course Title:** Software Engineering**Full Marks:** 50**Time:** 2 hours

Instruction: The figures in the right-hand margin indicate full marks for the questions. There are a total of 5 questions. Answer all the questions & missing data, if any, may be suitably assumed.

Q1(a). The team is beginning to design the application's data access layer, and two alternative approaches emerge, one prioritizing speed and another focusing on modularity. The team members have different opinions based on their past project experiences. Which team organization, democratic or chief programmer would lead to a more inclusive and well-analyzed decision in this situation, and why? [5 M]

Q1(b). Your company needs to develop a customer support mobile app within three months. The project requires developers, designers, and testers to work closely because features must be integrated and tested quickly. The client wants weekly updates and fast changes based on feedback. Currently, team members work in separate functional departments, which often causes communication delays. Which team organization structure functional, project, or matrix, would you choose for this fast-paced mobile app project, and why? [5 M]

Q2(a). Module M performs 12 operations. Out of these: 5 operations directly support the main task, 4 operations are loosely related helper tasks, and 3 operations are entirely unrelated. Then calculate the cohesion score. Identify whether cohesion is high/medium/low. [5 M]

Q2(b). The system contains 8 modules. Module M1 interacts with 5 other modules. M2 interacts with 3 modules. M3 interacts with 2 modules. The remaining modules (M4–M8) each interact with 1 module only. Then calculate the total number of inter-module interactions and coupling strength of the system. [5 M]

Q3. Identify the functional and non-functional requirements from the following Personal Finance Tracker system and document them. [10 M]

The Personal Finance Tracker System is designed to help users record and monitor their monthly spending across different areas of their daily life. The application will support up to ten expense categories, including food, travel, health, shopping, entertainment, utilities, education, investments, bills, and miscellaneous costs. Users can manage their expenses by adding new records, editing existing entries, or deleting any incorrect or outdated information. Each expense entry requires the user to specify a category code, the amount spent, the date of the transaction, and an optional description. When a new expense is added or modified, the system automatically updates the total monthly expenditure for the corresponding category and maintains an overall summary of all expenses. To make financial insights more visually understandable, the system provides a pie-chart view of the monthly expenditures, which users can toggle on or off using the view command. The Tracker will initially be developed as a desktop application but should be designed in a way that allows a future mobile version with cloud synchronization support, ensuring that users can access and back up their financial data across multiple devices. Throughout its operation, the system must remain lightweight, consuming minimal memory and processing power while still handling large volumes of entries efficiently. Additionally, it must enforce proper validation by alerting users when invalid

category codes are entered, non-numeric amounts are provided, required fields are left empty, or attempts are made to modify or remove entries that do not exist.

Q4. A college is organizing a National TechFest, and the project planning team has identified the major activities required to successfully conduct the event. The activities, their immediate predecessors, and estimated durations are given in Table-1. **[10 M]**

Table-1: Activities for TechFest Planning

Activity	Description	Immediate Predecessor	Duration (Days)
A	Finalize event theme	–	4
B	Prepare event posters and graphics	A	3
C	Create event website	A	5
D	Invite guest speakers	B	4
E	Open online registration	C, D	2
F	Arrange sponsorship meetings	A	6
G	Finalize sponsorship deals	F	3
H	Plan event schedule	E, G	4
I	Arrange logistics (stage, sound, lighting)	H	5
J	Conduct rehearsals	I	2
K	Execute TechFest	J	1

- Draw a Gantt chart for the activities, predecessors, and their durations in network diagram.
- Construct an activity network for this project and determine project duration and critical path.
- Determine Latest Start (LS) and Latest Finish (LF) of each activity in network diagram.
- Determine the Early Start (ES) and Early Finish (EF) of each activity in network diagram.
- Calculate Total Float (TF) and Free Float (FF) of each activity in network diagram.

Q5(a). A system accepts two input values and gives one output message based on the following rules: **[5 M]**

- The first input must be a vowel character (A, E, I, O, or U).
- The second input must be a positive integer (0–9).
- If both inputs satisfy the above conditions, the system prints: “Jab We Met”
- If the first input is not a vowel, the system prints: “Zindagi Na Milegi Dobara”
- If the second input is not a positive integer, the system prints: “Dil Chahta Hai”

Design the minimum number of required test cases for the above requirement specifications using the Cause–Effect Graph testing technique. Show all intermediate Cause–Effect graph and decision table.

Q5(b). Design the minimum number of test cases for the following requirement specifications using Equivalence Class Partitioning and Boundary Value Analysis techniques. **[5 M]**

A mobile payment application requires users to enter a 4-digit transaction PIN before confirming any payment. The PIN must satisfy the following conditions:

- It must contain exactly 4 digits.
- A PIN with less than 4 digits is invalid.
- A PIN with more than 4 digits is invalid.
- A PIN must contain digits only (0–9), no letters or special characters.
- Any invalid PIN should redirect the user to an “Invalid PIN” error screen.